

ASSIGNMENT 5 REPORT

TASKFLOW – PRODUCTIVITY DASHBOARD

[Github Link](#)

1. INTRODUCTION

Assignment 5 focused on developing a modern, dynamic, and responsive productivity management application using Flutter. Instead of creating a simple Todo application, the project was designed as a complete productivity dashboard named "TaskFlow".

The application provides users with a centralized interface to manage tasks, track productivity, organize projects and categories, view tasks on a calendar, and use a focus timer.

The main objective was to apply Flutter concepts such as responsive layouts, state management using StatefulWidget, reusable widgets, navigation, animations, forms, custom UI components, theme management, and interactive user interfaces.

The application was developed with a modern SaaS-style design using blue, cyan, and violet colors, rounded cards, icons, progress indicators, and responsive layouts.

2. OBJECTIVES

The main objectives of this assignment were:

- Build a professional productivity dashboard using Flutter.
- Create a responsive interface for desktop and mobile screen sizes.
- Implement dynamic task creation, completion, deletion, and search.
- Provide filtering for All, Active, and Completed tasks.
- Create separate screens for different productivity features.
- Implement navigation between multiple screens.
- Implement Light Mode and Dark Mode.
- Use reusable and modular Flutter widgets.
- Add animations and interactive UI elements.
- Create a focus/Pomodoro timer.
- Display productivity and task statistics dynamically.
- Maintain shared task data between related screens.
- Practice clean project organization using models, screens, widgets, and utility classes.

3. TECHNOLOGIES AND FLUTTER CONCEPTS USED

The following technologies and concepts were used:

- Flutter
- Dart
- Material 3
- StatefulWidget and State
- StatelessWidget
- ListView
- Row and Column
- Expanded and Flexible
- LayoutBuilder
- Responsive design
- NavigationBar
- CustomPaint and CustomPainter
- CircularProgressIndicator
- Forms and TextFormField
- showModalBottomSheet
- SnackBar
- Dialog
- AnimationController
- AnimatedContainer
- AnimatedSwitcher
- ScaleTransition
- Timer
- ThemeData
- Light/Dark themes
- Reusable widgets
- Model classes
- Git and GitHub

4. APPLICATION STRUCTURE

The project follows a modular structure:

```
lib/  
|  
|— main.dart  
|— app.dart  
|  
|— models/  
|   |— task_model.dart  
|  
|— screens/
```

```
| |— home_screen.dart
| |— tasks_screen.dart
| |— calendar_screen.dart
| |— projects_screen.dart
| |— categories_screen.dart
| |— focus_timer_screen.dart
|
|— utils/
|   |— app_colors.dart
|
|— widgets/
|   |— add_task_sheet.dart
|   |— filter_chips.dart
|   |— search_bar.dart
|   |— task_card.dart
```

The Task model is used to represent individual tasks. The screens folder contains the major application pages, while the widgets folder contains reusable UI components. App colors and styling constants are maintained separately for easier management.

5. MAIN FEATURES IMPLEMENTED

5.1 Dashboard

The Dashboard acts as the main productivity overview. It displays:

- Welcome section
- Total task count
- Active task count
- Completed task count
- Overall productivity percentage
- To Do List
- Productivity Tracker
- Quick actions

The dashboard values are calculated dynamically from the current task list. Therefore, completing or deleting tasks changes the displayed statistics.

5.2 Task Management

The Tasks screen provides a dedicated interface for managing tasks.

Users can:

- Create new tasks.
- Mark tasks as completed.
- Reopen completed tasks.
- Delete tasks.
- Undo deletion.
- Search tasks.
- Filter tasks by status.
- View total, active, and completed task counts.

The Add Task feature uses a modal bottom sheet containing a form with validation. The application also provides visual feedback using SnackBars.

5.3 Calendar

A dedicated Calendar screen was implemented to provide a calendar-based productivity interface.

It includes:

- Monthly calendar view
- Previous and next month navigation
- Today button
- Selected-date highlighting
- Daily agenda section
- Calendar legends
- Task-related agenda information

The screen provides a foundation for connecting tasks with dates in a future version.

5.4 Projects

The Projects screen provides a project-management style interface.

It includes several example projects such as:

- Flutter Assignment
- ML Learning
- Portfolio Website
- DSA Preparation
- Semester Planning
- Personal Goals

Each project displays information such as progress, task count, priority, and status. Category filters allow projects to be organized into different groups.

5.5 Categories

The Categories screen provides a visual overview of task organization.

The application contains categories such as:

- Development
- Learning
- Personal
- Work

The screen displays task distribution and completion statistics through professional cards and progress indicators.

5.6 Focus Timer

A Focus Timer was implemented using Dart's Timer functionality.

It supports:

- Focus mode
- Short Break mode
- Start
- Pause
- Reset
- Skip
- Multiple focus presets
- Completed session counter
- Daily session goal
- Circular progress indicator

The default focus duration is 25 minutes and the short break duration is 5 minutes.

6. RESPONSIVE DESIGN

Responsive design was an important part of the assignment.

The application detects the available screen width using `LayoutBuilder` and adapts the interface accordingly.

Desktop:

- Persistent sidebar

- Top navigation bar
- Multi-column dashboard layout
- Larger productivity panels
- Desktop-oriented task management interface

Mobile:

- Compact AppBar
- Bottom NavigationBar
- Responsive content
- Touch-friendly controls
- Mobile-friendly cards and spacing

This allows the same Flutter application to provide an appropriate experience across different screen sizes.

7. THEME AND VISUAL DESIGN

TaskFlow uses a modern productivity/SaaS visual style.

The interface uses:

- Blue as the primary color
- Cyan as a secondary accent
- Violet for additional highlights
- Green for successful/completed actions
- Orange for active/warning states
- Rounded cards
- Subtle borders
- Progress indicators
- Icons
- Gradients
- Consistent spacing

Both Light Mode and Dark Mode were implemented using separate ThemeData configurations.

Users can switch themes from the navigation interface without restarting the application.

8. ANIMATIONS AND USER INTERACTION

Animations were added to improve the user experience.

Examples include:

- Animated task completion controls
- Animated task card scaling
- Animated text style changes
- Animated containers
- Animated switchers
- Circular progress animations
- Interactive buttons and cards

Dialogs are used before deleting a task to prevent accidental deletion.

SnackBar provide feedback after important actions such as adding or deleting tasks.

9. SHARED APPLICATION STATE

The main HomeScreen maintains the application's task collection.

The same task data is passed to relevant screens such as:

- Dashboard
- Tasks
- Calendar
- Categories

This means that changes made to tasks are reflected in other parts of the application.

For example, when a task is completed from the Tasks screen, the completed task count and productivity percentage displayed on the Dashboard are updated.

10. TESTING AND VERIFICATION

The application was tested during development by running it in Flutter and checking the major functionality manually.

The following areas were verified:

- Dashboard loading
- Task creation
- Task completion
- Task deletion
- Undo deletion
- Search

- Task filters
- Dark/Light theme switching
- Sidebar navigation
- Mobile navigation
- Calendar interaction
- Project screen
- Categories screen
- Focus Timer controls
- Responsive desktop layout
- Responsive mobile layout

The project was also checked using Flutter analysis and the application was successfully run in the development environment.

11. CONCLUSION

Assignment 5 successfully expanded the basic Todo concept into a complete productivity dashboard application called TaskFlow.

The project demonstrates practical use of Flutter for building responsive, interactive, and visually polished applications. It combines task management, productivity tracking, project organization, calendar functionality, categories, and a focus timer into one application.

The assignment also provided practical experience with responsive UI, reusable components, state management, animations, themes, forms, timers, custom drawing, navigation, and Git/GitHub version control.

Overall, TaskFlow demonstrates how Flutter can be used to build a professional multi-screen application with dynamic functionality instead of a basic static interface.